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Comprehensive Psychiatric Assessment Service
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Clinical Neuropsychology

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Confidential: Report of Neuropsychological Evaluation

Name:	David Nimer
Referred by:	Jon Stevens, M.D.
Date of Birth (Age):	5/31/1991 (24)
Dates of Testing:	2/2/2016
ID Number:	245053

BACKGROUND INFORMATION: Mr. Nimer is a 24-year-old male who was referred for neuropsychological assessment of his cognitive functioning as part of his ongoing outpatient evaluation at the Menninger Clinic. Psychological assessment is being conducted by another provider. His detailed history is fully documented elsewhere and will not be reiterated; for further detail, please see his psychosocial and psychiatric evaluations, as well as detailed timelines provided by his parents. Chief concerns relate to instances of problematic/imprudent behavior and poor judgment, particularly in the aftermath of a significant head injury in February 2015. His parents raise concern that mood and behavioral problems may have worsened further in the aftermath of another accident and October 2015 and may have contributed to multiple subsequent citations for driving under the influence of a substance. Available records include the report from prior evaluations for ADHD. The evaluation conducted in March 2011 was not found to support that diagnosis, while evaluation conducted in June 2014 did support that diagnosis. Both reports also referenced the presence of behavioral traits including impulsivity, difficulty delaying gratification, reduced sensitivity to behavioral consequences, poor judgment, skepticism, irritability, and a tendency toward defiance/rebelliousness. The 2014 report also references Mr. Nimer's substance use.

At the time of the current evaluation, Mr. Nimer reported that he continues to have ongoing difficulties with organization and completion of paperwork, but denied having any history of hyperactivity or inattention. He expressed frustration that he was unable to capitalize on multiple leads at work because he did not manage or complete needed paperwork. He speculated that his inherent difficulties with organization may have gone unchecked because he was able to perform well throughout much of his education without meeting to be organized or applying maximal effort. He reported that prior treatment with Adderall improved his focus and academic performance, but that he discontinued the medication approximately one month ago due to concerns that taking the medication would negatively influence his chances should he apply to join the military. He reported that in the aftermath of his head injury he experienced increased difficulty with memory, concentration, and management of his time schedule, but believes he has experienced significant recovery. He described his baseline level of intellectual functioning is very high and expressed some uncertainty as to whether he has returned to his full baseline. He added that since the October 2015 accident his mood has been "harsher" and more intolerant.

Behavioral Observations: Mr. Nimer completed this assessment over the course of one day with breaks taken as needed. His mood appeared euthymic. He appeared to provide his history in a forthright, open,

and engaged manner. He exhibited good social skills, although times he presented as somewhat cocky. He reported that he considered "gaming" the tasks and was confident that he would be able to do so, but then decided against that tact. He appeared to provide good effort during the cognitive assessment, often exhibiting a high level of persistence. He had a competitive style and appeared driven to display his intellectual capacity. In the face of task difficulty, he nevertheless remained engaged and did not attempt to provide exculpatory rationales or act dismissively toward the tasks. Spontaneous speech was fluent, coherent, and very articulate for his age. Language comprehension was fully intact. He required fairly minimal repetition of directions. Environmental testing conditions were adequate and the following is considered to provide a valid estimate of his neuropsychological functioning at the time of the assessment.

TESTS ADMINISTERED: Orientation, Wechsler Test of Adult Reading, Wechsler Adult Intelligence Scale – Fourth Edition (WAIS-IV), Test of Memory Malingering, Rey Auditory Verbal Learning Test, Wechsler Memory Scale-Fourth Edition (Logical Memory), Brief Visuospatial Memory Test-Revised, Clock Drawing, intersecting pentagons, cube copy, ramparts, Rey Osterrieth Complex Figure Test, Trail Making Test, Verbal Series Attention Test, Stroop Color and Word Test, Wisconsin Card Sorting Test, Conners' Continuous Performance Test - II, Boston Naming Test, Lexical Fluency, Category Fluency, ASRSv1.1.

TEST RESULTS: [NOTE: In the results section, the following phrases describe performance within the range of percentiles indicated in the parentheses: Severely Deficient (0.6 or lower); Deficient (0.7 – 1.9); Impaired (2 – 9); Low Average (10 – 24); Average (25 – 74); High Average (75 – 90); Superior (91 – 97); Very Superior (98 or higher)]

Effort: On a measure of effort, administered in the guise of a visual memory test his performance was adequate on the initial trial (45/50) and approached ceiling level on the second trial (49/50). The subtle error on the latter trial appeared to be secondary to an instance when he misspoke when providing his response, rather than a genuine error. Following a delay, his performance was at ceiling level (50/50). This pattern of performance is indicative of adequate effort. His subsequent performances and behavioral observations provided further indication of optimal effort.

Orientation: He was fully oriented.

Intellectual Functioning: He was administered a measure of general intellectual functioning and obtained a Full Scale IQ score of 119, which is at the upper limit of the high average range (90th percentile). However, the meaningfulness of this aggregate score is limited due to significant scatter among his index and subtest scores. His performance profile reflected a relative strength for Verbal Comprehension, as he obtained an associated index score in the very superior range. This, in part, reflected a peak subtest score in the very superior range for word knowledge. He obtained index scores in the high average range for Working Memory and Perceptual Reasoning. He exhibited a relative weakness for Processing Speed, obtaining an associated index score in the average range. His General Abilities index score, which discounts the influence of attention and processing speed, was 127, which is in the superior range (96th percentile). The obtained subtest scores in the average range for working memory span, code transcription speed, and speeded visual matching. All other subtest scores were in the high average range or above, including general fund of knowledge, which was in the superior range. The following IQ, scale, and individual subtest scores were obtained on the WAIS-IV:

	<u>Standard Score</u>
Full Scale IQ	119 (limited meaning)
Verbal Comprehension Scale	132
Perceptual Reasoning Scale	117
Working Memory Scale	111
Processing Speed Scale	97

<u>Verbal Comprehension</u>	<u>Scaled Score</u>	<u>Perceptual Reasoning</u>	<u>Scaled Score</u>
Vocabulary	19	Block Design	14
Similarities	13	Matrix Reasoning	12
Information	14	Visual Puzzles	13
<u>Working Memory</u>	<u>Scaled Score</u>	<u>Processing Speed</u>	<u>Scaled Score</u>
Digit Span	9	Coding	8
Arithmetic	15	Symbol Search	11

Memory: Immediate recall of two prose passages was in the high average range. After a delay, his recall of these passages was in the superior range and reflected a retention rate of 100% for learned information. His accuracy on yes/no questions regarding the passages' content was in the high average range. On a word list-learning task, his initial recall was in the average range (6/15 words). His recall more than doubled on the second trial, yielding a score in the superior range (13/15 words). His recall was high average to superior on the three subsequent learning trials (14, 15, and 15/15 words, respectively). His recall of a distracter list was in the average range (7/15 words). Following both this distraction and a longer delay his recall remained in the superior range and at ceiling level (15/15 words for both). His initial immediate recall of an array of geometric designs was in the impaired range. His recall more than doubled on the second trial, improving to the average range. On the third and final learning trial, his recall was high average and at ceiling level. Following a delay, his recall remained high average and at ceiling level.

Language Functioning: Naming to confrontation was in the average range for his age (54/60) and improved further with the provision of phonemic cues (56/60). Verbal fluency for semantic categories was in the very superior range. As referenced above, his ability to define a series of words was in the very superior range and reflected a relative strength. Reading/decoding of a series of single words was in the superior range.

Visuoperceptual Skills: His copies of intersecting pentagons and a cube were accurate. He was able to copy and extrapolate a repeating pattern. He was also able to draw a clock depicting an instructed time. Visuoconstruction with blocks was in the superior range; he was able to complete all items correctly and often rapidly. His copy of a complex geometric figure scored significantly below expectation (30.5/36 points); the majority of the figure was adequately represented, but he omitted two significant elements from the bottom portion of the design. When subsequently asked to review his output he was readily able to detect this omission.

Attention and Executive Functions: Repetition span was maximally six digits forward, which is low average. Working memory span was maximally five digits backward and six digits numerically sequenced, which are average. His total score across these digit span tasks was average. As noted above, speeded visual search/matching and speed of code transcription were average. On a graphomotor task requiring visual scanning and mental tracking, his time to completion was in the average range for both a single sequence and a more complex alternating sequence; he made no errors on either trial. Mental tracking for familiar verbal material was in the superior range for speed and was completed without error. Mental arithmetic was superior; he was able to solve many of the problems quite rapidly. Mental integration of fragmented designs, verbal abstract reasoning, and nonverbal deductive reasoning were high average. Lexical fluency was also high average. On a measure of sustained attention and impulse control, his overall performance was classified as "good/satisfactory;" although his response speed was somewhat slow and varied in reaction to different stimuli presentation rates and he made a slightly elevated number of commission errors (18/36). On a reasoning task requiring mental flexibility, concept formation, and utilization of performance feedback, he completed four of six possible sets, which is in

the impaired range. His accuracy rate was low average and he made an elevated number of loss of set errors. His vocalizations throughout this task indicated that he was attempting to utilize an overly complex set of hierarchical rules that tended to derail him from consistently utilizing more on point observations and inferences.

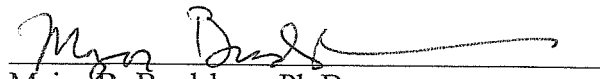
Attention Self-Report: On a symptom checklist assessing for the presence and frequency of symptoms associated with ADHD, he reported experiencing 3 of 6 core symptoms and 6 of 12 supplementary symptoms with significant frequency. This level of endorsement is below the threshold suggestive of significant problems with attention and restlessness in daily life.

Behavioral Rating Self and Informant Reports: On a behavioral rating scale emphasizing skills associated with executive functioning, his endorsements were indicative of significant difficulties with inhibition, task monitoring, and planning/organization. On the same inventory, completed by his mother, her responses were suggestive of broad difficulties with behavioral regulation, impacting inhibition, set shifting, emotional control, and self-monitoring, as well as perceived difficulties with working memory, planning/organization, and task monitoring. A comparison of the two inventories suggests that his mother perceives greater and broader behavioral difficulty than he does; greatest discrepancy was seen for set shifting and emotional regulation.

SUMMARY AND CONCLUSIONS: On cognitive testing Mr. Nimer's performance was almost exclusively in the average range or above and included numerous scores in the high average to very superior range. This was reflected in his obtained WAIS-IV General Ability index score of 127, which is in the superior range (96th percentile). He exhibited exceptionally strong verbal skills, including superior fund of semantic knowledge and word reading and very superior semantic fluency and breadth of vocabulary. Demonstrating for well-rounded abilities, he also scored in the superior range on select measures of visuoconstruction, mental arithmetic, speeded verbal inhibition, and memory. Contrasting with these areas of excellence, he exhibited mild relative weakness on select measures of attention span, sustained attention, speed of processing, and initial recall on rote learning tasks. It is also noted to make a careless omission error when copying a complex design. His performance on these measures was often in the low average or merely average range, which for him is suggestive of relative difficulty. Additional difficulty was seen on a measure of concept formation, mental flexibility, and deductive reasoning, but this appeared secondary to "overthinking" the task.

This pattern of performance is suggestive of a relative weakness in attention, which arises in the context of otherwise generally high average to very superior intellectual abilities. This pattern of performance could be accounted for by reported elements of Attention Deficit Disorder; he provided a history suggestive of chronic problems with organization, prioritization, and impulsivity. His strong intellectual endowment may have allowed him to mask or compensate for other common symptoms/manifestations. The untoward impact of his history of brain injury, as well as substance use, should be considered as additional potential exacerbating contributory factors. This is particular the case regarding the former because his parents provide a compelling timeline of worsening function in the aftermath of his head injuries, particularly as environmental demands increased. A comparison of available data from his 2011 evaluation and the current assessment revealed significant improvements for verbal abstract reasoning, word knowledge, nonverbal deductive reasoning, and mental arithmetic. No instances of decline were observed. This finding is somewhat surprising given the intervening head injuries, but may reflect the beneficial effects of his recent period of sobriety. These noted improvements also curb concern regarding a significant persistent cognitive sequela associated with TBI. However, it is noted that cognitive sequela and behavioral sequela did not always tightly correlate. Given his excellent

performance on testing, areas of relative difficulty notwithstanding, his recent behavioral difficulties cannot be explained by cognitive factors in isolation.


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